Atul Boyal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.06
SHIKSHA HIGH SCHOOL SHIVBARI BIKANER RAJ	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2024	91.00%
LYALL PUB SCH BHOJANSHALA GHARSISAR BIKANER RAJ	Class X (CENTRAL BOARD OF SECONDARY EDUCATION)	2022	97.20%

TECHNICAL SKILLS

Languages: C++, Python, C, JavaScript, TypeScript, SQL, HTML, CSS, Assembly (RISC-V)

Databases & Backend: PostgreSQL, Supabase

Frameworks & Libraries: React.js, Node.js, Tailwind CSS

Systems & Technologies: Linux, xv6, RISC-V, eBPF, ARM NEON

Developer Tools: Git, GitHub, QEMU

PROJECTS

TaskVault | Full-Stack Realtime Productivity Platform GitHub | Live

- Developed an end-to-end productivity platform using React, Supabase, and PostgreSQL with recurring habits, analytics, Focus Mode, and cross-device synchronization
- Built a multi-client ecosystem spanning a React PWA and Chrome extension with optimistic UI updates, Supabase Realtime synchronization, device-local vault authentication, and PostgreSQL Row-Level Security

RepoInsight-AI | AI-Powered Repository Analysis Platform GitHub | Live

- Built an AI-powered repository analysis platform using React, TypeScript, Express.js, and Gemini, combining repository traversal, static code analysis, and metadata extraction to generate structured engineering reports
- Engineered a multi-stage processing pipeline with smart source-code filtering, context construction, response validation, progress tracking, and session-based logging for reliable and efficient repository analysis
- Integrated Gemini for AI-assisted code review and implemented structured validation of LLM-generated responses before incorporating them into engineering reports.

Hardware-Aware Hybrid Task Scheduler

- Built a Linux scheduler using eBPF, PMU telemetry (**perf_event_open**), and CPU affinity to classify workloads and dynamically optimize task placement across heterogeneous P/E cores

- Classified workloads using IPC, floating-point activity, and sleep/runtime ratio, with EWMA-based smoothing to stabilize runtime decisions and prevent migration thrashing
- Reduced energy consumption by **11.7%** and average power draw by **27.1%** for memory-bound workloads through workload-aware task placement

Extending xv6 | Operating Systems & Kernel Development GitHub

- Extended the xv6 RISC-V kernel with custom system calls, per-process accounting, and a 4-level System-Call-Aware MLFQ scheduler featuring interactive workload detection and starvation prevention
- Built a virtual memory subsystem implementing Clock page replacement, swapping, frame management, and priority-aware page eviction under memory pressure

Accelerating Integer Computations Using Underutilized Floating-Point Units

- Implemented and benchmarked ARM NEON SIMD optimizations on Raspberry Pi 5 across matrix multiplication, integer dot product, image convolution, K-Means, and merge sort workloads.
- Performed **assembly-level analysis** to exploit **SIMD** and **instruction-level parallelism (ILP)** beyond compiler auto-vectorization
- Achieved up to **2.68× speedup** over compiler -O3 optimized implementations while maintaining **100% output correctness** across compute, memory, and branch-bound workloads

EXPERIENCE

Defen.do

Project Contributor - Frontend Development & Containerized Lab Infrastructure

- Developed reusable frontend components and interactive workflows for courses, modules, lessons, quizzes, and practical cybersecurity labs using React and TypeScript
- Contributed to browser-accessible practical lab functionality involving Docker-based isolated Parrot OS containers and VNC/noVNC technologies

ACHIEVEMENTS

- **Codeforces Specialist** with a maximum rating of **1459**
- Secured **1st Place at Lambda Hackathon 2025, IIT Hyderabad** as part of Team Helios

RELEVANT COURSEWORK

Computer Science: Data Structures & Algorithms, Computer Architecture, Operating Systems, Database Management Systems, Theory of Computation, Compilers, Artificial Intelligence, Software Development Fundamentals

Mathematics: Probability & Random Variables, Statistics, Linear Algebra, Calculus, Transform Techniques

POSITION OF RESPONSIBILITY

Sunshine, IIT Hyderabad GitHub

Web & App Development Team Member

- Built the new Sunshine web application from scratch using React as the sole web developer for the first three months, developing reusable, data-driven components and responsive interfaces to establish a maintainable and consistent frontend codebase